

Peizheng Li

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Summary

Ph.D. candidate and research engineer specializing in **foundation models for autonomous driving** at the intersection of **VLM/VLA systems, world models, and 3D scene understanding**. Research focus includes multimodal LLMs for driving, spatial reasoning, open-world perception for long-tail scenarios, automated dense pseudo-label generation over large-scale multi-sensor data, and open-/closed-loop evaluation. First-author publications at **CVPR, ICCV, and IJCAI**; experienced in building industrial-scale training, evaluation, and data pipelines across cloud and local GPU systems.

Work Experience

- Mercedes-Benz Group AG**, Stuttgart, Germany *Feb. 2023 - Present*
Industrial Ph.D. Researcher, Scene Understanding Group, RD
- Conduct research on foundation models for autonomous systems, spanning **VLM/VLA for autonomous driving, open-world 3D occupancy prediction, spatial/world-model-oriented reasoning**, diffusion-based motion planning, **self-supervised scene flow**, and selected **robotics / HRI** collaborations.
 - SpaceDrive** (CVPR 2026, 1st author): end-to-end spatial VLM that fuses uniform 3D positional encoding with vision-language tokens and integrates chain-of-thought reasoning; outperforms prior VLA baselines on nuScenes open-loop planning and the Bench2Drive closed-loop benchmark.
 - AGO** (ICCV 2025, 1st author): VLM-guided open-world 3D occupancy prediction via adaptive cross-modal grounding, enabling **zero-shot novel-class recognition** and rapid few-shot transfer for long-tail robustness.
 - Built large-scale data-engine and evaluation pipelines: data acquisition, cleaning, temporal consistency, scene / instance retrieval, and **automated dense 3D pseudo-label generation (auto-labeling)** over multi-sensor driving datasets.
 - Scaled training and evaluation across local GPU clusters and cloud environments using **PyTorch, CUDA, DDP, Docker, Kubernetes, Flyte, Azure, and GCP**.
 - Mentored interns and master thesis students; collaborated across research and engineering teams on open-source releases and reproducible evaluation.
- Mercedes-Benz Group AG**, Stuttgart, Germany *Apr. 2022 - Dec. 2022*
Master Student, Scene Understanding Group, RD
- Built a camera-based end-to-end BEV perception and future prediction model with Lift-Splat-Shoot-style view transformation, temporal prediction, and multi-task decoding; achieved **39.3% dynamic IoU** on nuScenes.
 - Designed a multi-stage temporal GCN for the Waymo Occupancy and Flow Prediction Challenge; achieved **74.2% Flow-Grounded Occupancy AUC** and ranked **4th** on the leaderboard.
- Mercedes-Benz Group AG**, Stuttgart, Germany *May 2021 - Oct. 2021*
Research Intern, Scene Understanding Group, RD
- Studied contextual bias in 2D object detection and domain adaptation; showed that disruptive context reduced mAP by **5.37%** on benchmark experiments.
 - Developed context-separation components and baseline pipelines to support internal adaptation research.

Publications

- SpaceDrive: Infusing Spatial Awareness into VLM-based Autonomous Driving** *CVPR 2026*
Peizheng Li, Zhenghao Zhang, David Holtz, Hang Yu, Yutong Yang, Yuzhi Lai, Rui Song, Andreas Geiger, Andreas Zell
- AGO: Adaptive Grounding for Open World 3D Occupancy Prediction** *ICCV 2025*
Peizheng Li, Shuxiao Ding, You Zhou, Qingwen Zhang, Onat Inak, Larissa Triess, Niklas Hanselmann, Marius Cordts, Andreas Zell
- SeFlow: A Self-Supervised Scene Flow Method in Autonomous Driving** *ECCV 2024*
Qingwen Zhang, Yi Yang, Peizheng Li, Olov Andersson, Patric Jensfelt
- PowerBEV: A Powerful Yet Lightweight Framework for Instance Prediction in Bird's-Eye View** *IJCAI 2023*
Peizheng Li, Shuxiao Ding, Xieyuanli Chen, Niklas Hanselmann, Marius Cordts, Juergen Gall
- G2DP: Diffusion Planning with Spatio-Temporal Grid Guidance** *IROS 2026*
Hang Yu, Ye Jin, Alessandro Canevaro, Julian Jordan, Peizheng Li, Marc Kaufeld, Silvan Lindner, Johannes Betz, Wilhelm Stork
- Shift & Drift: A Zero-Shot Benchmark for Generalizable and Robust Autonomous Driving Motion Planning**
Alessandro Canevaro, Hang Yu, Peizheng Li, Silvan Lindner, Wilhelm Stork, Georg Martius, Julian Jordan *IROS 2026*

TQD-Track: Temporal Query Denoising for 3D Multi-Object Tracking

BMVC 2026

Shuxiao Ding, Yutong Yang, Julian Wiederer, Markus Braun, Peizheng Li, Juergen Gall, Bin Yang

FAM-HRI: Foundation-Model Assisted Multi-Modal Human-Robot Interaction Combining Gaze and Speech

Yuzhi Lai, Shenghai Yuan, Boya Zhang, Benjamin Kiefer, Peizheng Li, Andreas Zell

T-ASE 2026

Sticky-Glance: Robust Intent Recognition for Human Robot Collaboration via Single-Glance

arXiv 2026

Yuzhi Lai, Shenghai Yuan, Peizheng Li, Andreas Zell

SEER-VAR: Semantic Egocentric Environment Reasoner for Vehicle Augmented Reality

arXiv 2025

Yuzhi Lai, Shenghai Yuan, Peizheng Li, Jun Lou, Andreas Zell

Education

University of Tübingen

Feb. 2023 – Present

Ph.D. Candidate in Computer Science

◦ Advisors: **Prof. Andreas Geiger** and **Prof. Andreas Zell**

◦ Research: **multimodal learning, VLM/VLA, world models, embodied AI, robotics and autonomous driving**

University of Stuttgart

Apr. 2020 – Dec. 2022

M.Sc. in Electromobility

◦ Grade: **1.6** (German scale, **1.0** best; approx. **3.7/4.0** US equivalent)

◦ Thesis: *End-to-End Agent Perception and Occupancy-Based Motion Prediction in Bird's-Eye View*

Tongji University

Sept. 2014 – Jul. 2019

B.Eng. in Vehicle Engineering

◦ Grade: **4.5 / 5.0**

◦ Thesis: *Vehicle Detection and Tracking Based on Sensor Fusion*

Technical Skills

Programming: Python, C++, CUDA, C#, MATLAB, Bash

ML / Systems: PyTorch, TensorFlow, OpenMMLab, Distributed Training, Docker, Kubernetes, Flyte, W&B, Azure, GCP

Research Areas: foundation models & VLM/VLA for driving, world models, multimodal learning, 3D vision, open-world perception, BEV / occupancy, scene flow, auto-labeling & data engines, spatial intelligence, embodied AI, multi-view geometry, SLAM, HRI

Languages: Chinese (native), English (professional), German (professional)

Service

Reviewer: CVPR 2025/2026, ICCV 2025, ECCV 2026, ICRA 2026, AAAI 2026/2027, IROS 2025/2026, IEEE T-ITS 2025

Organizer: 4th & 6th DriveX Workshops, in conjunction with CVPR 2026 & ECCV 2026